MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 6952

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Beam power tetrode, pulse amplifier
See figure 1
Mounting position: Tube axis vertical, either end up
Weight: 38 pounds (17.24 kg) nominal

ABSOLUTE RATINGS: F = 600 MHz

Parameter: / Unit:	Ef Vac (Note 1)	eb kv (Note 2)	ec2 kv (Note 3)	Ecl Vdc	ib a	Ib mAdc	icl a (Note 5)	Icl mAdc	ic2 a
Anode pulse amp class B									
tp = 15 µs(note 4)									
Maximum:	1.00(dc) 1.00(60 Hz) 1.05(400 Hz)	55	2.2	-400	80	320	15	60	15
Minimum:	---	---	---	---	---	---	---	---	---
tp = 70 µs(note 4)									
Maximum:	1.00(dc) 1.00(60 Hz) 1.05(400 Hz)	30	2.2	-400	30	500	3	60	3
Minimum:	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	0.95(60 Hz)	---	---	---	---	---	---	---	---

ABSOLUTE RATINGS: F = 600 MHz

Parameter: / Unit:	Ic2 mAdc	Pi kW	Pp kW	tk sec (Note 6)	Cooling --- (Note 7)	Environmental requirements --- (Note 9)	Operating frequency MHz	Fault protection --- (Note 10)	Duty ---	If(surge) A (Note 6)
Anode pulse amp class B										
tp = 15 µs(note 4)										
Maximum:	60	16	8	---	---	---	174-600	---	0.004	550
Minimum:	---	---	---	90	---	---	---	---	---	---
tp = 70 µs(note 4)										
Maximum:	60	9	5	---	---	---	174-600	---	0.02	550
Minimum:	---	---	---	90	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	Note 8	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
Ⓓ	Qualification inspection								
---	Vibration	23, 24		---	---	---	---	---	---
---	Vibration - test end point:								
---	Peak power output (1)	-		---	---	po	750	---	kw
Ⓓ 1042	Shock, specified pulse	23, 25	Test condition A	---	---	---	---	---	---
---	Shock-test end point:								
---	Peak power output (1)	-		---	---	po	750	---	kw
	Quality conformance inspection, part 1								
Ⓓ 1155	Anode water pressure differential	14	No voltages applied	0.65	II	APD	---	5.0	psi
Ⓓ 1155	Grid (1) water pressure differential	14	No voltages applied	0.65	II	GPD	---	6.0	psi
Ⓓ 1155	Grid (2) water pressure differential	14	No voltages applied	0.65	II	SPD	---	8.0	psi
Ⓓ 1155	Filament water pressure differential	14	No voltages applied	0.65	II	FPD	---	8.0	psi
Ⓓ 1155	Cathode water pressure differential	14	No voltages applied	0.65	II	CPD	---	8.0	psi
1301	Filament current	8	Ef and cooling only	0.65	II	If	460	530	Aac
---	Emission check	17	Ef/If = 450 Aac, 60 Hz; eb = 30 kv; ec1, or Ec1 (see note 16); ec2 = 2 kv (max); Rf drive/Ib = 165 mAdc (max); Ic1 = 0 to +28 mAdc; F = 225 ± 15 MHz; tp = 10 ± 1 µs; prr = 300 ± 5	0.65	II	po	825	---	kw
---	Peak power output (2)	17	eb = 30 kv; Ec1 (see note 16); ec2 = 2 kv (max); Rf drive = 5 kw (max); Pd/Ib = 0.6 Adc; output circuit bandwidth = 5 MHz (min); t = 60 minutes; tp = 60 µs (min); duty = 0.018; F = 400 to 550 MHz	0.65	II	po	300	---	kw
1256	Electrode current (grid)	18, 20	Eb = 6.5 kVdc; Ec2 = 600 Vdc; Ec1/Ib = 1.0 Adc; t = 5 minutes	0.65	II	Ic1	---	-50	mAdc
1256	Electrode current (screen)	-	Ec2 = 600 Vdc; Ec1/Ib = 1.0 Adc; Eb = 6.5 kVdc	0.65	II	Ic2	-25	+25	mAdc
---	Input resonant frequency	21	No voltages applied	---	---	Fi	222	250	MHz
---	Output resonant frequency	21	No voltages applied	---	---	Fo	230	250	MHz

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2								
1331	Direct-interelectrode capacitance	22	Use special shield adapter DESC Drawing No. 66060	---	---	Cg2k Cg1p	0.01 ---	0.018 0.15	µF pF
Ⓓ1261	Electrode voltage (grid 1)	15	eb = 26 kv; ec1 or Ec1/Ib = 8.0 mAdc; Ef = 0.95 V, 60 Hz; ec2 = 2 kv	0.65	II	ec1 or Ec1	---	-320	v or Vdc
Ⓓ---	Peak power output (1)	17	eb = 30 kv; ec1 or Ec1 (see note 16); ec2 = 2 kv (max); RF drive/Ib = 165 mAdc; Ic1 = 0 to +28 mAdc; Ic2 = 45 mAdc (max); F = 225 ±15 MHz; t = 60 minutes; tp = 10±1 µs; prr = 300±5	0.65	II	po	825	---	kw
Ⓓ---	Power gain	17,19	Peak power output (1); except po = 825 kw (min)	---	---	PG	100	---	---
	Quality conformance inspection, part 3								
---	Life-test group S	-		---	---	---	---	---	---
---	Initial life	1,11, 13,17	Peak power output (1); po = 750 kw (min); ec2 = 2 kv (max); specified life = 1,000 hours	---	---	---	---	---	---
---	Life-test end point:								
---	Peak power output (1)	-		---	---	po	750	---	kw
---	Multiple short life	12	Same as initial life except po = 750 kw (min)	---	---	t	50	---	hrs
---	Life-test end point:								
---	Peak power output (1)	-		---	---	po	825	---	kw
Ⓓ---	Sequential life	1,13, 17	Eb = 15 kVdc; ec2 = 2 kv (max); Ec1 (see note 16); Pd/Ib = 0.81 Adc; Po = 250 kW (min); F = 400 MHz (min); tp = 60 µs (min); prr = 300±5; t = 1,000 hours	---	---	---	---	---	---
---	Life-test end point:								
---	Sequential life-test conditions	-		---	---	po	750	---	kw

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NOTES:

1. Because the filament voltage, when operated near the maximum value, provides emission in excess of any requirements within tube rating, during life the filament voltage shall be reduced to a value that will give adequate, but not excessive emission. Careful attention to maintaining the value consistent with adequate emission will result in conserving the life of the tube. The filament voltage shall be measured at the respective liquid coolant connections on the tube side of the threads. This procedure is essential in order to accurately measure the filament voltage. At 400 Hz some heating of the filament leads and rf cathode terminal (cathode header) occurs; this condition is not detrimental to tube operation or tube life.

2. Peak positive anode voltage. The peak pulse value is defined as the maximum value of a straight line through the average of the fluctuations over the top portion of the pulse. The magnitude of any spike, on the anode voltage pulse, shall not exceed the peak pulse value by more than 4,000 volts, and the duration of any spike (measured at a point 70% above the peak pulse value) shall not exceed 2 µs. In the absence of rf output voltage, the peak pulse voltage may be exceeded by 20 percent but not more than 10 consecutive pulses. A pressurized output cavity is required, for operation at the maximum voltage specified herein, to prevent external flashover at the insulating bushing. (See note 9.)

3. Peak positive screen voltage. The peak pulse value is defined as the maximum value of a straight line through the average of the fluctuations over the top portions of the pulse. The magnitude of any spike, on the grid(2) voltage pulse, shall not exceed the peak pulse by more than 250 volts and the duration of any spike (measured at a point 70% above the peak pulse value) shall not exceed 2 µs.

4. On time (tp) is defined as the sum of the durations of all the individual pulses which occur during any 3,000 µs interval.

5. Peak rectified grid(1) current.

6. Minimum time to reach operating filament voltage shall be 30 seconds. After operating filament voltage has been obtained, a minimum of 90 seconds heating time is required before other electrode voltages can be applied. Filament surge current shall not exceed 550 amperes, even momentarily.

7. Cooling requirements.

   (a) Air cooling. Forced-air cooling of the ceramic bushing at the grid(1) seal and at the anode seal may be required in order to limit the temperature of the ceramic bushing at either seal to the specified maximum value of 150°C. Under such conditions, provisions shall be made for blowing air at the ceramic bushing through suitable openings in the coaxial-cylinder cavity circuit.

   (b) Liquid cooling. Liquid cooling of the filament block, dc cathode block, grid(1) block, grid(2) block, and anode, is required. When tube operation under low-ambient temperatures is required, the recommended coolant is inert liquid FC 75[1/], or equivalent, but ethylene glycol mixed with water in the proportion of 60 percent ethylene glycol to 40 percent water by weight can be used. When the environmental temperature is above 0°C, the coolant may be water. The liquid flow shall start before application of any voltages and preferably shall continue for several seconds after removal of all voltages. Interlocking of the liquid flow through each of the cooled elements with all power supplies is recommended to prevent tube damage in case of failure of adequate liquid flow. Liquid flow shall be as shown in table I.

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7. -Continued

TABLE I. Liquid flow.

	Absolute minimum flow (gpm)	Typical flow (gpm)	Maximum pressure differental 2/ for typical flow (psi)
With water:			
Through filament block	0.5	0.8	8
Through dc cathode block	0.5	0.8	8
Through grid(1) block	0.5	0.8	6
Through grid(2) block	0.5	0.8	8
Through anode in direction shown on figure 1:			
For anode dissipation up to 5 kw (average)	5	7	5
For anode dissipation of 8 kw (average)	8	10	10
With inert liquid FC 75, or equivalent: (with outlet liquid temperature of 30°C)			
Through filament block	1.0	1.2	20
Through dc cathode block	1.0	1.2	20
Through grid(1) block	1.0	1.2	14
Through grid(2) block	1.0	1.2	20
Through anode in direction shown on figure 1:			
For anode dissipation up to 5 kw (average)	10	12	20
For anode dissipation of 8 kw (average)	20	24	80
With ethylene-glycol-water solution:			
Through filament block	1.0	1.2	18
Through dc cathode block	1.0	1.2	18
Through grid(1) block	1.0	1.2	12
Through grid(2) block	1.0	1.2	18
Through anode in direction shown on figure 1:			
For anode dissipation up to 5 kw (average)	6	8	7
For anode dissipation of 8 kw (average)	16	18	40

1/ Made by the Fluorochemical Division, Minnesota Mining and Manufacturing Company.

2/ Directly across cooled element.

8. The minimum waterflow to the anode cooling connection shall be 5 gpm. The minimum waterflow to the filament, filament-cathode, grid(1), and grid(2) cooling connections, shall be 0.5 gpm. Waterflow shall start before the application of any voltages, and shall be continued for at least 6 seconds after removal of all voltages. The outlet water temperature shall not exceed 70°C.

Emergency conditions: Simultaneous interruption of cooling pump and tube electrode power is permissible.

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NOTES: -Continued

9. Environmental requirements.

External gas pressure (absolute) 1/	60 2/ psi (max)
Gage pressure at any inlet	100 psi (max)
Metal-surface temperature	100°C (max)
Ceramic-bushing temperature	150°C (max)
Outlet liquid temperature (any outlet) for water and FC 75, or equivalent	70°C (max)
Outlet screen temperature (any outlet) for ethylene glycol	60°C (max)
Storage temperature 3/	-65°C (min)

1/ Absolute pressure is the sum of the atmospheric pressure and gage pressure.
2/ This pressure is related to the output cavity pressurization as required to prevent corona or external flashover.
3/ The tube coolant ducts shall be free of liquid before storage or shipment of the tube to prevent damage from freezing, except that storage at -65°C is permissible with FC 75 coolant, or equivalent installed, and at -45°C with ethylene-glycol solution installed.

10. High-speed electronic protective devices shall be used to prevent energy from the anode-power supply, grid(1) power supply, and from the grid(2) power supply from permanently damaging the tube and its circuitry in the event of abnormal operation. A suitable high-speed electronic protective device shall be capable of deactivating the various pulse sources supplying the tube during the inter-pulse interval following the occurence of a fault in any single pulse, and shall be capable of meeting the requirements of the aluminum foil simulated fault tests. Thus, all pulses subsequent to a "fault" pulse shall not be applied until the device is reset. (A fault is a condition in which the peak pulse currents rise 20 percent or more above the normal peak current level.) Additional protection against damage from an anode-to-grid(2) fault shall be provided to prevent the grid(2) voltage from rising to a value in excess of 3,000 volts during the fault pulse. Minimum reset time shall be 1 second.

11. This requirement shall be waived for manufacturers qualified to MIL-E-1/1106.

12. The following sampling plan shall be used:

Lot size	Sample	Sample size	Cumulative sample size	See 10.1.2 of MIL-STD-105 Accept	Reject
1	First	1	1	0	1
	Second	0	0	-	-
2	First	1	1	0	-
	Second	1	2	1	2
3 to 10	First	1	1	0	-
	Second	2	3	1	2
Over 10	First	3	3	0	2
	Second	3	6	1	2
	Second (see note)	3	6	2	3

NOTE: The second sample criteria applies only if the first sample from the previous lot had zero failures.

13. Water system requirements. In order to minimize maintenance, it is recommended that the water cooling system be of the closed type, utilizing water of acceptable quality as specified herein, to prevent excessive deposition, corrosion, and electrolysis. Deposition of material not only restricts waterflow, but also prevents proper transfer of heat from the tube elements to the cooling water. Corrosion and electrolysis may destroy the tube elements, ducts, and fittings. Cooling water of acceptable quality shall have a specific resistivity in excess of 1 megohm-cm.

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NOTES: - Continued

14. The waterflow to the anode cooling connection shall be 7 gpm. The waterflow to the filament, filament-cathode, grid(1) and grid(2) cooling connections, shall be 0.8 gpm.

15. The anode and grid(2) voltage pulse length (tp) shall be 10 ± 1 μs with a repetition rate of 300 ± 5 Hz.

16. The grid bias voltage is adjusted for class B operation. Class B operation is defined as a negative grid voltage which is between one-fifth and one-eight of the ec2 voltage.

17. Circuit adjustments are permissible. The rectified rf output pulse length (tp) and the pulse repetition rate (prr) shall be as specified herein. Peak output power is calculated by dividing the average output by the duty factor. The average output power is measured by a calorimetric method using a dummy water load, RCA Model MI 19198-A, or equivalent. The specified test duration (t) shall be for continuous operation.

18. This test is to be the first test performed at the conclusion of the holding period.

19. Power gain is the quotient of the peak-output power and the peak rf input power. Peak rf input power shall be calculated by dividing the average rf input power by the Du. Average rf input power shall be measured with a calibrated directional coupler. The average rf input power is the difference between the forward and reverse values as indicated by the directional coupler. Calibration of the directional coupler shall be accomplished by the calorimetric method, or equivalent.

20. With conditions as specified herein, grid(1) current (I_1) shall be recorded after specified time. All voltages shall be removed except filament voltage and grid(1) voltage, and then grid(1) leakage current (I_2) shall be recorded. Calculate Icl from the formula: $Icl = I_1 - I_2$.

21. The following procedure (or equivalent) shall be used for measuring input and output resonant frequency.

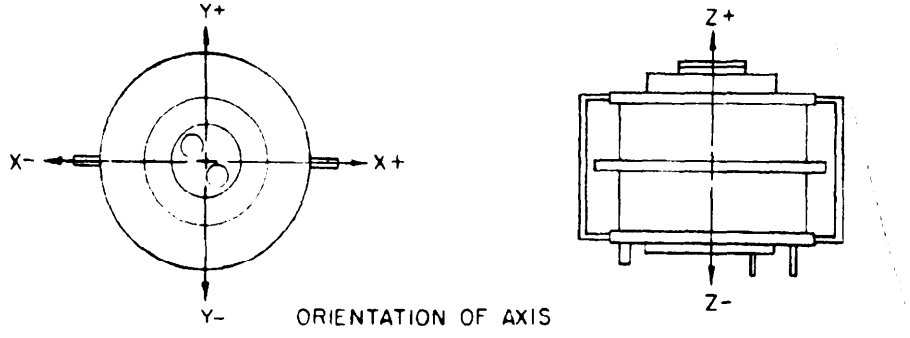
    (a) Set up equipment as in figure 2. Be sure tube is making good contact with cavity.

    (b) Adjust the frequency of rf generator to obtain a resonant peak on the standing-wave indicator.

    (c) The resonant frequency as read on frequency meter shall be within limits specified herein. (The input cavity and output cavity shall be as shown on figures 3 and 4.)

22. Each element not under test shall be tied to ground. Two tubes per month shall be tested. Failure of one or more tubes will require 100-percent testing. Two tubes per month may be resumed after testing of 20 consecutive tubes without failure. Any tube having one or more interelectrode capacitances greater than 100 pF shall be measured for all of its capacitances on suitable capacitance or impedance bridges operating at 1,000 Hz.

23. Vibration and shock tests shall be performed using vibration-shock fixture (DESC Drawing No. 67031). Tests shall be performed with no voltages applied.



ORIENTATION OF AXIS

Vibration and shock-test end points need only be performed once (at the conclusion of both vibration and shock tests).

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NOTES: - Continued

24. Vibration.

(a) The tube shall be vibrated in the X, Y, and Z axes at the following conditions:

Frequency (Hz)	Double amplitude (inches)
5 to 55	0.060

(b) Frequency shall be swept logarithmically from 5 to 55 to 5 Hz in 5 minutes minimum. Two sweeps shall be made.

(c) The tube shall be vibrated in the X, Y, and Z axes at the following conditions:

Frequency (Hz)	Double amplitude (inches)	Duration (hours/axis)
50	0.060	12

(d) Tubes which do not comply with the post-vibration limits shall be considered failures.

(e) This is not a destructive test.

25. Shock. Test in accordance with MIL-STD-202, method 213, test condition A, except as follows:

(a) The peak acceleration of the shock shall be 15G ± 10 percent.

(b) The velocity change (V_j) = 3.39 ft/sec.

(c) Tubes which do not comply with post-shock limits shall be considered failures.

(d) This is not a destructive test.

Custodians:
Army - EL
Navy - Ec
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2998)

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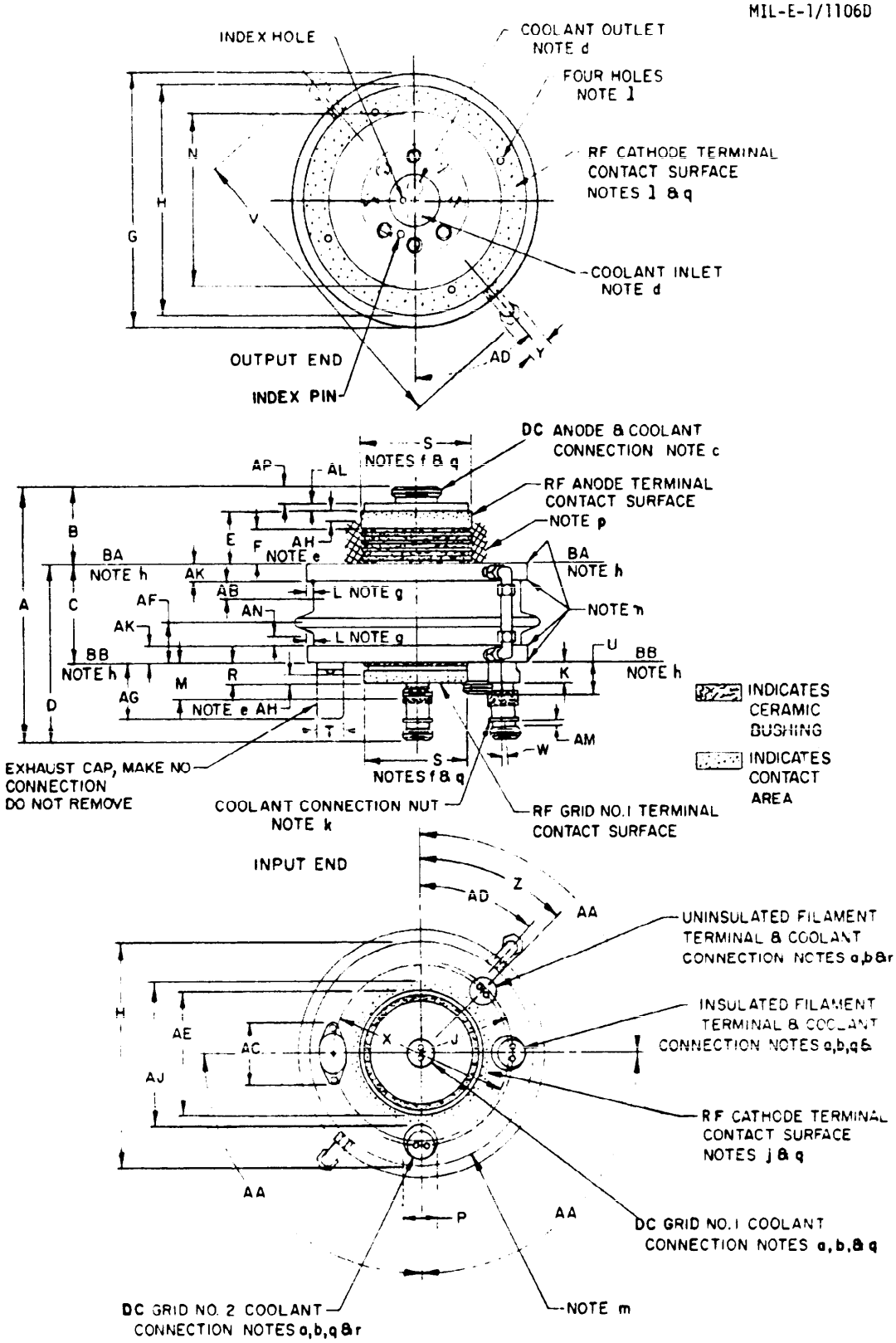


FIGURE 1. Outline drawing of electron tube type 6952

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			
	G-3 gage check		q
E	1.83 (46.48)	1.93 (49.02)	
R	.59 (14.99)	.69 (17.53)	
S	3.74 (95.00)	3.76 (95.50)	f, q
Quality conformance inspection, part 2			
N		6.00 (152.40)	l
P	1.22 (30.99)	1.28 (32.51)	
V		11.25 (285.75)	
X	6.19 (157.23)	6.31 (160.27)	q
AB	.50 (12.70)		g
AC	2.09 (53.09)	2.15 (54.61)	
AE		4.06 (103.12)	j
AF	1.38 (35.05)	1.50 (38.11)	
AH	.38 (9.65)		e
AJ	4.95 (125.73)		j
AN	.25 (6.35)		g
Quality conformance inspection, part 3			s
A	8.31 (211.07)	8.93 (226.82)	
C	3.27 (83.06)	3.45 (87.63)	
D	5.68 (144.27)	6.18 (156.97)	
F	1.17 (29.72)		
G		8.75 (222.25)	
H	7.84 (199.13)	7.88 (200.15)	
K		.59 (14.99)	
L	.13 (3.30)		g
T	.97 (24.64)	1.12 (25.45)	
U	.83 (21.08)	.95 (24.13)	
Y		.68 (17.27)	
AG		2.00 (50.80)	
AK	.56 (14.22)	.63 (16.00)	
AL		.28 (7.11)	
AM	.09 (2.29)		
AP	.52 (13.21)	.60 (15.24)	
	Reference dimensions		
B	2.69 (66.33)		q
M	1.09 (27.69)		q
J	3.25 (82.55) Rad.		q
W	3°		q
Z	45°		q
AA	90°		q
AD	41°		q

FIGURE 1. Outline drawing of electron tube type 6952 - Continued.

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NOTES:

a. Terminal has 1 inch (25.4)–16 unified thread, class 2A fit, .380-inch (9.65) long, and two holes .258 – .270-inch (6.55–6.86) diameter, spaced .438 inch (11.13) on centers. Quality conformance inspection, part 1, test.

b. The holes in the filament, grid No. 1, and grid No. 2 coolant terminal connections, will accept the pins of the plug-and-cylinder combination gage G1, shown on figure 5. Quality conformance inspection, part 1, test.

c. Terminal has 1-3/4 inch (44.45)–16 unified extra fine thread, class 2A fit, .380-inch (9.65) long, two holes .508–.522-inch (12.90–13.26) diameter, spaced .688 inch (17.48) on centers, and an index hole .160-inch (4.06) maximum diameter, spaced .344 inch (8.74) from the center of the terminal. Quality conformance inspection, part 1, test.

d. The holes in the anode coolant connection shall accept the pins of the plug-and-cylinder combination gage G2, shown on figure 6. Quality conformance inspection, part 1, test.

e. Pressure from circuit contacts shall be exerted only over maximum length, AH, of designated contact areas of the anode or grid No. 1 terminals.

f. This diameter dimension is held only over a length of AH.

g. Dimension L applies over length AB and AN, as indicated.

h. The contact surfaces, BA-BA and BB-BB of the rf cathode terminals, shall be parallel within .06 inch (1.52).

j. Contact of the input end, rf cathode terminal, shall not be made at a diameter smaller than 4.06 inch (103.12), nor greater than 4.95 inch (125.73).

k. To prevent excessive stress on the ceramic seal, a 15/16-inch (23.81) open-end wrench shall be used to permit gripping the terminal when removing or tightening the coolant connectors.

ℓ. Contact of the output end, rf cathode terminal, shall not be made at a diameter smaller than 6 inches (152.40). The pressure exerted for this rf contact, shall be limited to that necessary for good electrical contact. The mechanical force for the cavity support and pressure seal shall be made at a diameter not less than 6 inches (152.40). On the output end, rf cathode terminal, there are four equally spaced .188-inch (4.78) diameter holes on a circle having a diameter of 6.75 inches (171.45). These holes are for tube manufacturing purposes only. Attention is called to the existence of these holes so that equipment designers can avoid making a pressure seal or electrical contact at points which are coincident with these holes. Mechanical clamping devices for the output cavity shall be designed so as to exert their clamping force across outer edge of output-header flange.

m. Serial number is located on this surface between grid No. 2 and the insulated filament terminal.

n. Corners may be rounded or chamfered, but not to exceed .05 inch (1.27).

p. Keep this 5-inch diameter annular-volume region clear. Do not allow circuit components (conductors or insulators) to protrude into this region.

q. The output end shall accept the output end of gage G3, shown on figure 7. Proper acceptance is obtained when gage is seated on the output end, rf cathode terminal, contact surface. The gage is properly seated when a .015-inch (.38) thickness gage, 1/2-inch (12.70) wide will not enter between the gage and the rf cathode contact surface. The pins on the 1.500-inch (38.10) movable section of the output end of gage G3, shall enter the anode coolant holes such that the height slot represented by the 2.098-inch (53.29) dimensions is below the shoulder of the gage, and the 2.219-inch (56.36) dimension is visible above the shoulder. The input end shall accept the input end of gage G3, shown on figure 7. Proper acceptance is obtained when the gage is seated on the input end, rf cathode terminal surface, and the projecting guide pins of the output end engage the mating holes of the input end. The gage is properly seated when a .015-inch (.38) thickness gage, 1/2-inch (12.70) wide will not enter between the gage and the rf cathode terminal contact surface. The pins on the .875-inch (22.23) movable section of the input end shall enter the grid coolant holes such that the height slot represented by the 2.379-inch (60.43) dimension is below the shoulder of the gage, and the 2.559-inch (65.00) dimension is visible above the shoulder.
When gage G3 is properly seated:
(1) Dimensions B, J, M, W, X, Z, AA, and AD shall vary only within the limits of the gage.
(2) The centerline through the grid No. 1 coolant connection holes will have a maximum angle of 2 degrees with respect to the centerline of the anode coolant connection holes.
(3) The axis of the rf anode terminal contact surface is coincident with the axis of the grid No. 1 terminal contact surface, within .05 inch (1.27).
(4) The axis of the anode and coolant connection is coincident with the axis of the rf anode terminal contact surface, within .025 inch (.64).

r. The centers of the water course holes, in each connection around the periphery of the tube, shall be on a line through the center of these holes and perpendicular to a line through the center of the tube within ±2 degrees.

Ⓓs. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 6952 - Continued.

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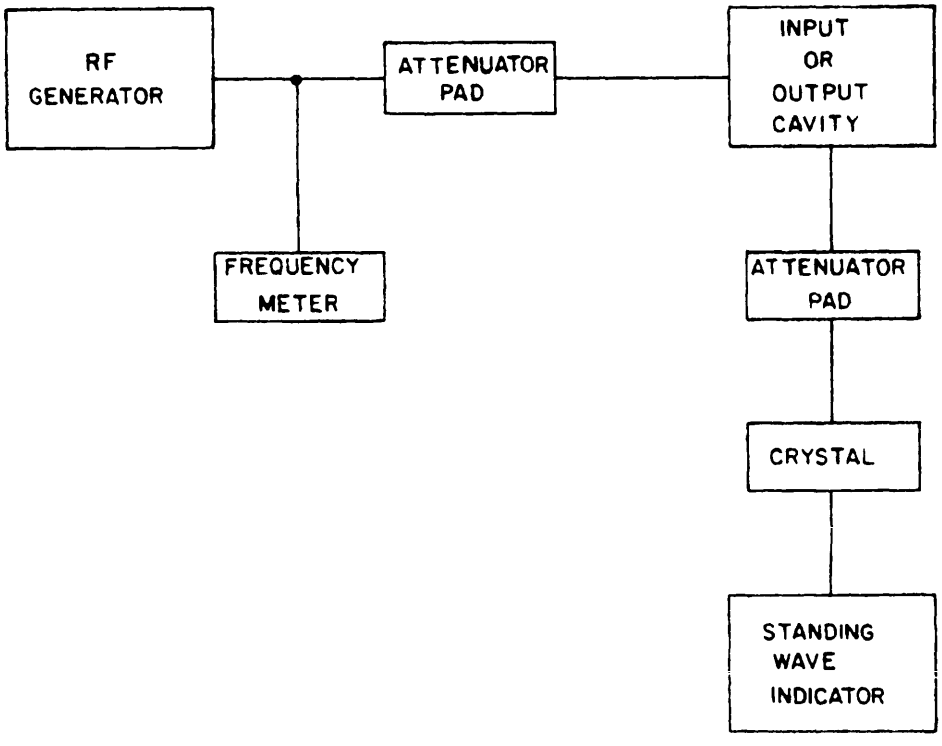
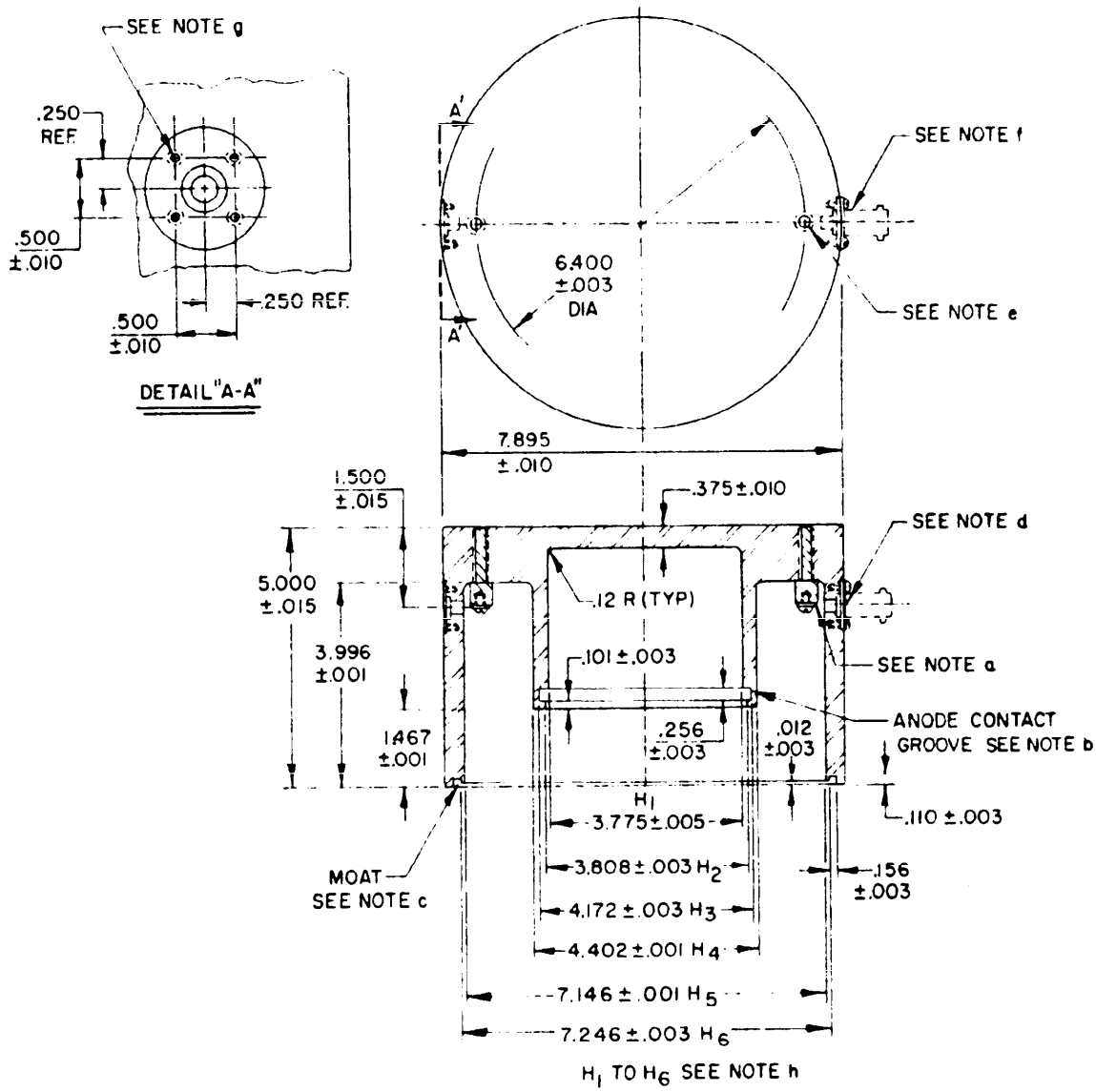


FIGURE 2. Typical block diagram of test circuit.

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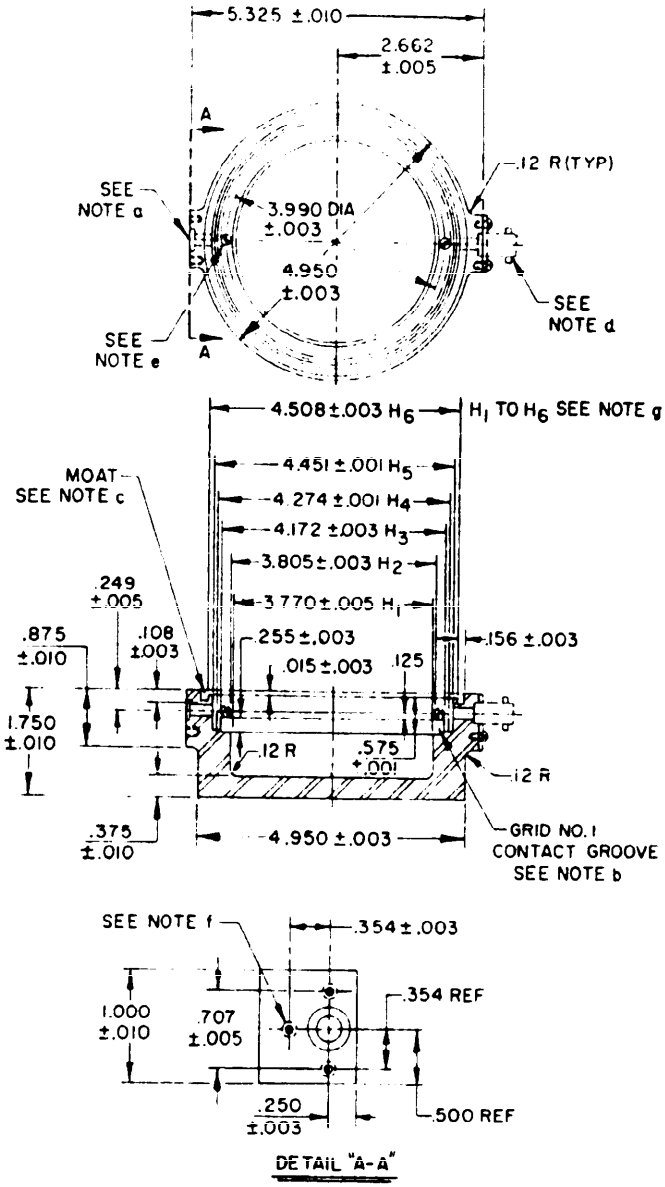


NOTES:

a. Two studs. 1-1/2-inch long threaded 1/4-inch-20 NC by 1 inch with 3/8-inch-diameter head. Head of stud shall be drilled and tapped 1/4-inch deep for No. 6-32 RH machine screw. Use standard 5/32-inch-flat washer under screw.
b. Anode contact groove shall be filled with the following spring: .020-inch-diameter wire wound to give a .250-inch O.D. at 35 TPI, 14-1/4 inches in length (approx.).
c. Moat shall be filled with a braid to insure continuous braid contact and yet permit firm seating of the cavity on the rf cathode terminal contact surface.
d. Two holes. Drill 7/32-inch diameter (thru), C' bore 3/8- diameter by 1/8-inch deep. Spot face 1-inch diameter by 1/16-inch deep.
e. Two holes. Drill and tap for 1/4-inch-20 NC screws.
f. Location of A-N connector-UG-290 U (two connectors 180 degrees apart).
g. Four holes. Drill and tap for No. 3-48 screws by 1/4-inch deep.
h. Cylindrical holes H_1 to H_6 have axes coincident within .001 inch.

FIGURE 3. Output cavity required for resonant frequency test.

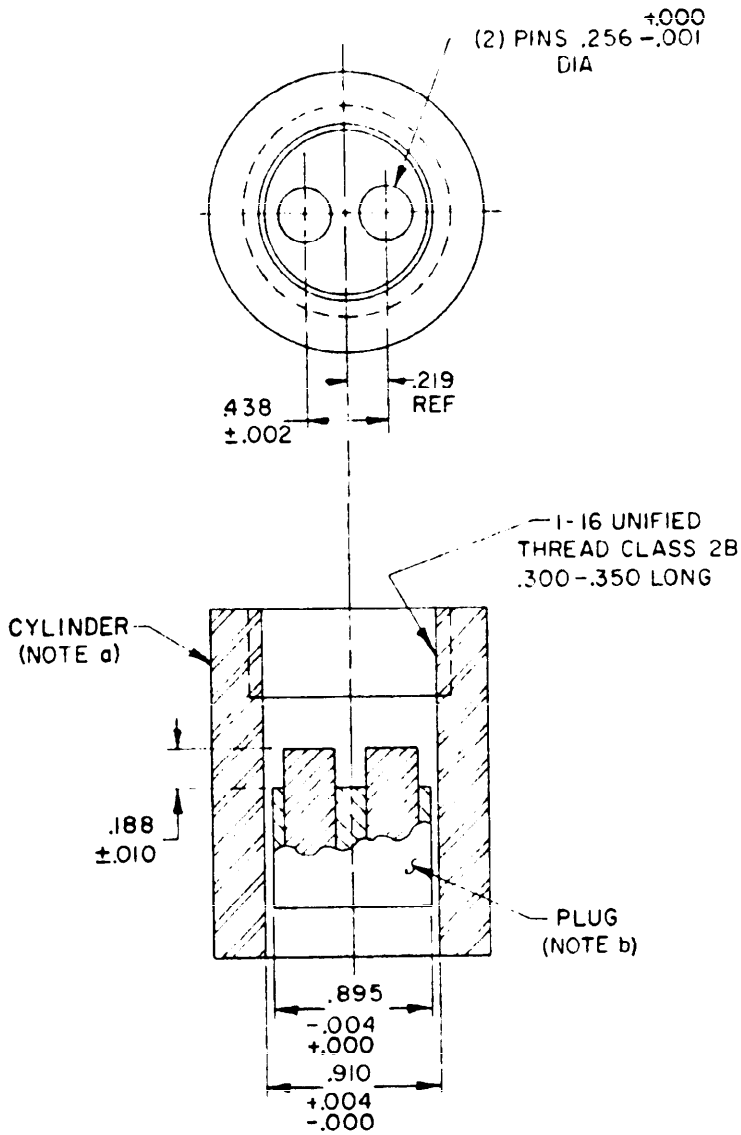
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NOTES:
a. Two holes, drill 7/32-inch diameter (thru), C' bore 3/8-inch diameter by 1/16-inch deep.
b. Grid No. 1 contact groove to be filled with the following spring: .020-inch-diameter wire wound to give a .250-inch O.D. at 35 TPI, 14-1/4 inches in length (approx.).
c. Moat shall be filled with a braid to insure continuous braid contact and yet permit firm seating of the cavity on the rf cathode terminal contact surface.
d. Location of A-N connector - UG-290/U (two connectors 180 degrees apart).
e. Two holes. Drill and tap for No. 3-48 screws.
f. Three holes. Drill and tap for No. 3-48 screws by 3/16-inch deep.
g. Cylindrical holes H_1 to H_6 have axes coincident within .001 inch.

FIGURE 4. Input cavity required for resonant frequency test.

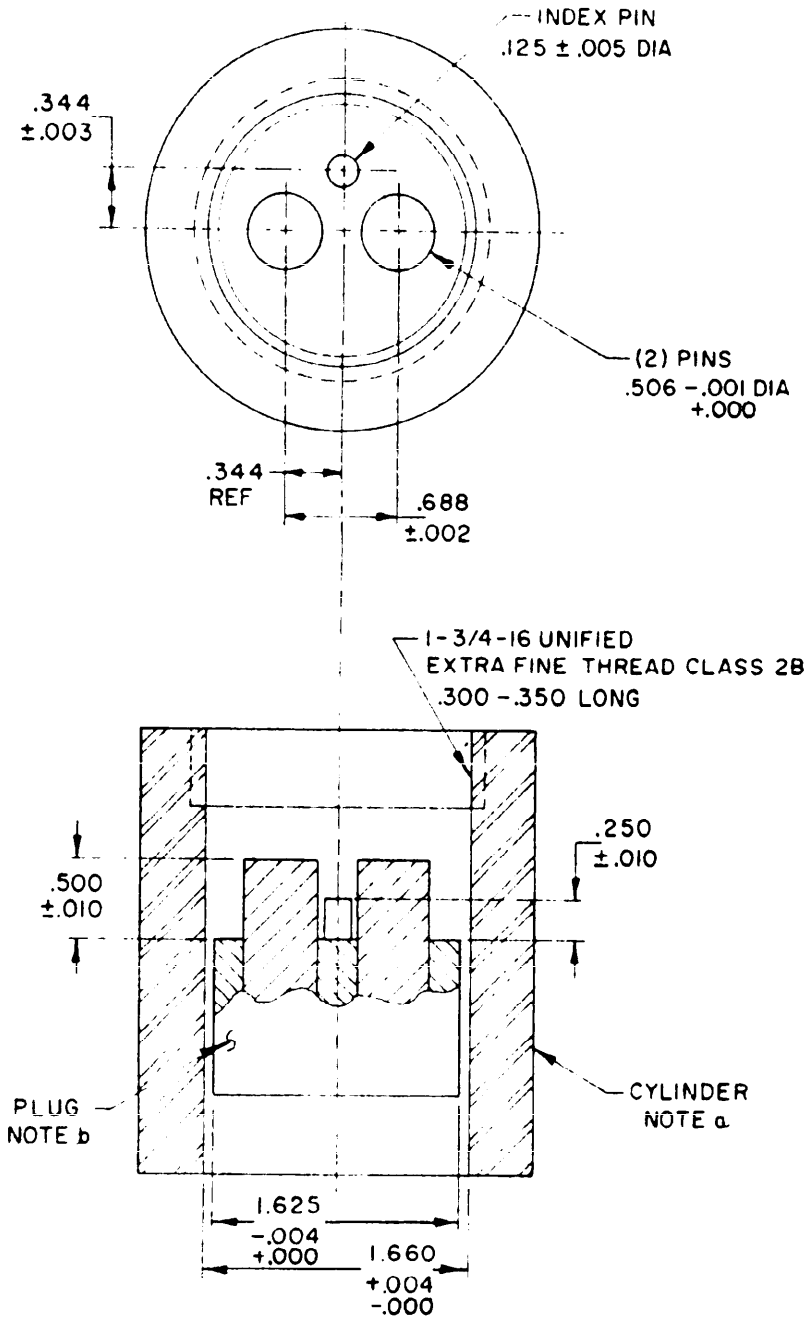
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NOTES:
a. Axis of the cylindrical holes and threaded section shall be coincident within .002 inch.
b. Plug sides and pin sides shall be parallel within .001 inch.

FIGURE 5. Gage G-1.

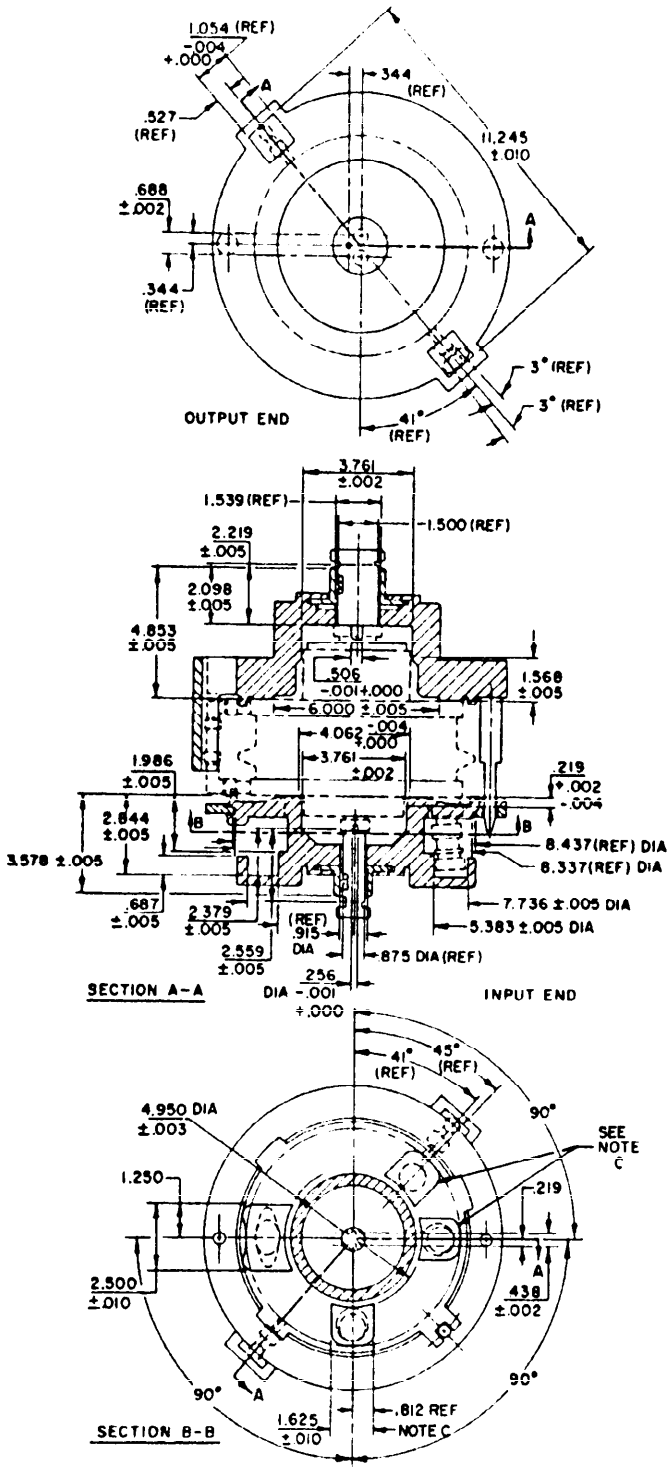
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NOTES:
a. Axis of the cylinder holes and threaded section shall be coincident within .002 inch.
b. Plug sides and pin sides shall be parallel within .001 inch.

FIGURE 6. Gage G-2.

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NOTES:

a. Tolerances shown are gage tolerances and shall not be considered tube tolerances.
b. See DESC drawing No. 67038.
c. Dimensions apply for all three slots.

FIGURE 7. Gage G-3.

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